



21 DAYS SQL CHALLENGE

DAY 17

BY AMRUHA AHMED

DAILY CHALLENGE

Create a report showing each service with: service name, total patients admitted, the difference between their total admissions and the average admissions across all services, and a rank indicator ('Above Average', 'Average', 'Below Average'). Order by total patients admitted descending.

```
SELECT
service,
SUM(patients_admitted) AS "Total Patients Admitted",
ABS(SUM(patients_admitted) - ROUND((SELECT AVG(total) FROM (SELECT SUM(patients_admitted) AS total
FROM services_weekly GROUP BY service) AS d1)))AS "Difference",
CASE
WHEN SUM(patients_admitted)> (SELECT AVG(total) FROM (SELECT SUM(patients_admitted) AS total FROM
services_weekly GROUP BY service) AS d3) THEN "Above Average"
WHEN SUM(patients_admitted)< (SELECT AVG(total) FROM (SELECT SUM(patients_admitted) AS total FROM
services_weekly GROUP BY service) AS d4) THEN "Below Average"
ELSE "Average"
END AS "Category"
FROM services_weekly
GROUP BY service;
```

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OUTPUT

service	Total Patients Admitted	Difference	Category
emergency	1185	278	Below Average
surgery	1686	223	Above Average
general_medicine	2332	869	Above Average
ICU	648	815	Below Average

PRACTICE QUESTION 1

Show each patient with their service's average satisfaction as an additional column.

```
SELECT
patient_id AS "ID",
p.service AS "service",
ROUND(avg_patient_satisfaction,2) AS "Average Patient Satisfaction"
FROM patients p JOIN
(SELECT AVG(patient_satisfaction) AS avg_patient_satisfaction , service FROM services_weekly GROUP BY
service) AS s
ON p.service=s.service;
```

ID	service	Average Patient Satisfaction
PAT-09484753	surgery	79.27
PAT-f0644084	surgery	79.27
PAT-ac6162e4	general_me...	81.23
PAT-3dda2bb5	emergency	77.88

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PRACTICE QUESTION 2

Create a derived table of service statistics and query from it

```
SELECT * FROM  
(SELECT  
  ROUND(AVG(available_beds)) AS "Average Available Beds",  
  ROUND(AVG(patients_request)) AS "Average Patients Request",  
  ROUND(AVG(patients_admitted)) AS "Average Patients Admitted",  
  ROUND(AVG(patients_refused)) AS "Average Patients Refused",  
  ROUND(AVG(patient_satisfaction)) AS "Average Patient Satisfaction",  
  service  
FROM services_weekly  
GROUP BY service) AS stats;
```

Average Available Beds	Average Patients Request	Average Patients Admitted	Average Patients Refused	Average Patient Satisfaction	service
23	119	23	96	78	emergency
38	43	32	11	79	surgery
46	82	45	37	81	general_medicine
15	15	12	3	82	ICU

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PRACTICE QUESTION 3

Display staff with their service's total patient count as a calculated field.

```
SELECT staff_id AS "ID",  
s.service AS "service",  
patients_count AS "Total Patients"  
FROM  
staff s INNER JOIN  
(SELECT service, COUNT(patient_id) AS patients_count FROM patients GROUP BY service) AS p  
ON s.service=p.service;
```

ID	service	Total Patients
STF-b19d1900	surgery	254
STF-5d51a8ea	surgery	254
STF-1ded4330	surgery	254
STF-42844552	surgery	254

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The background features two abstract, flowing liquid-like shapes on the left and right sides. These shapes are composed of vibrant, iridescent colors including blue, green, yellow, and orange, creating a sense of movement and depth against the solid black background.

THANK YOU

for your time and attention

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